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(54) **SPORTS PADDLE**

(71) Applicant: **Mitch Junkins**, Newport Beach, CA (US)

(72) Inventor: **Mitch Junkins**, Newport Beach, CA (US)

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(60) Provisional application No. 63/339,855, filed on May 9, 2022, provisional application No. 63/419,154, filed on Oct. 25, 2022, provisional application No. 63/754,416, filed on Feb. 5, 2025.

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(52) **U.S. Cl.**

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(57) **ABSTRACT**

An apparatus and methods are provided for a sports paddle for pickleball, tennis, smashball, ping pong, or racquetball. The sports paddle comprises a planar portion for striking a ball and a handle portion for grasping in a hand. A skin layer encloses the planar portion and the handle portion. Rounded edges are disposed around the planar portion and the handle portion. The planar portion may comprise an internal string core coupled with structured foam layers and surrounded by carbon fiber layers. The planar portion may comprise a core material surrounded by a rigid portion. The rigid portion may comprise 20-24 lb. foam while the core material is a honeycomb structured material. A synthetic or polymeric grip is wrapped around the handle portion to provide stability and firmness for a player. A binding secures the grip to a neck of the planar portion.

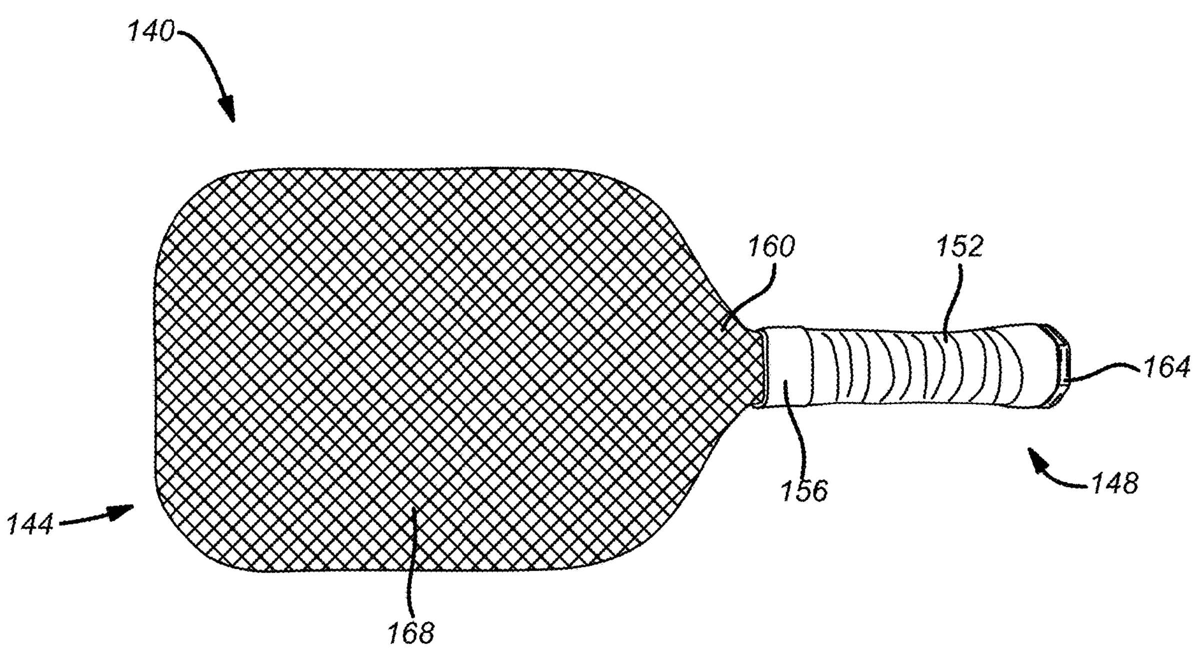
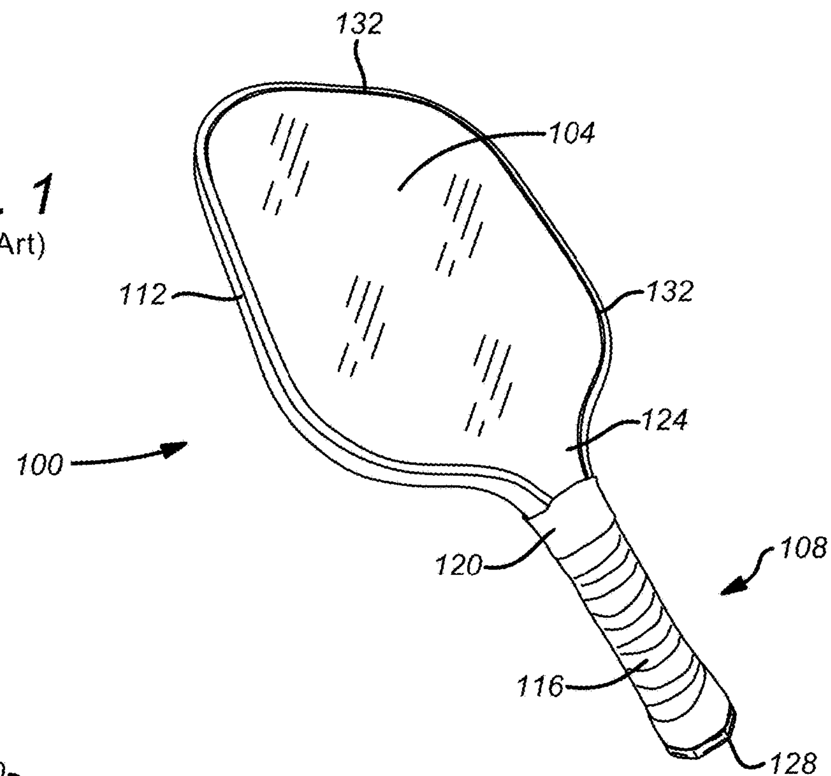


FIG. 1
(Prior Art)



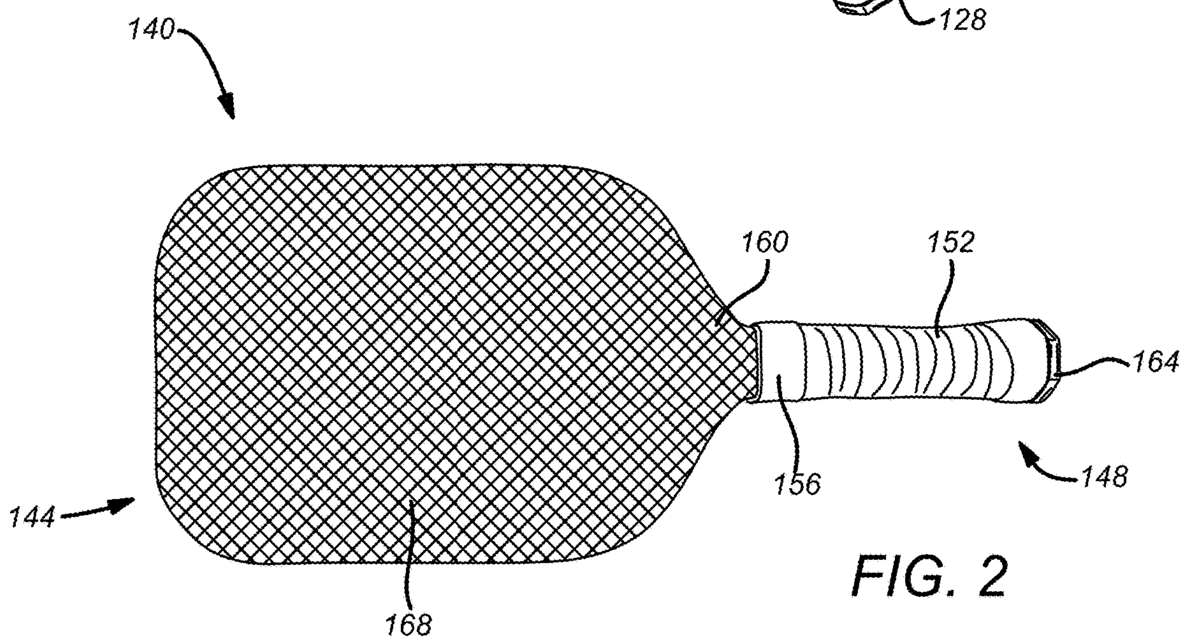
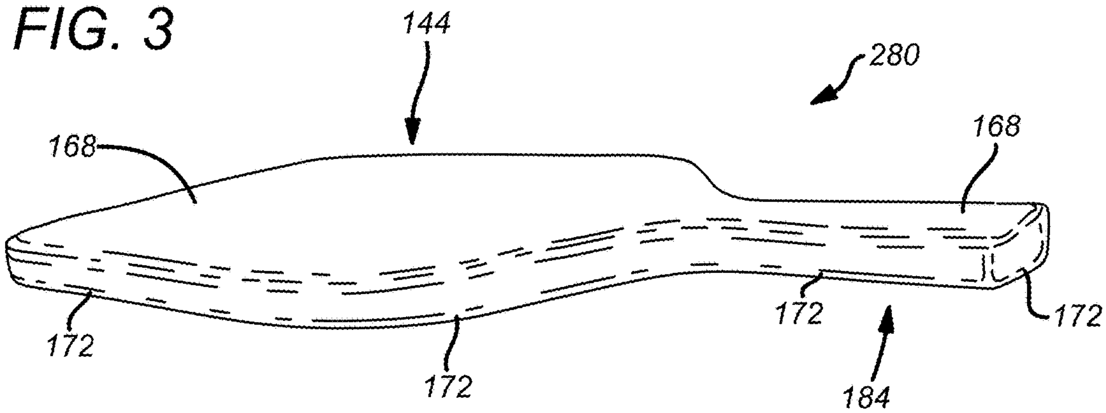




FIG. 2

FIG. 3



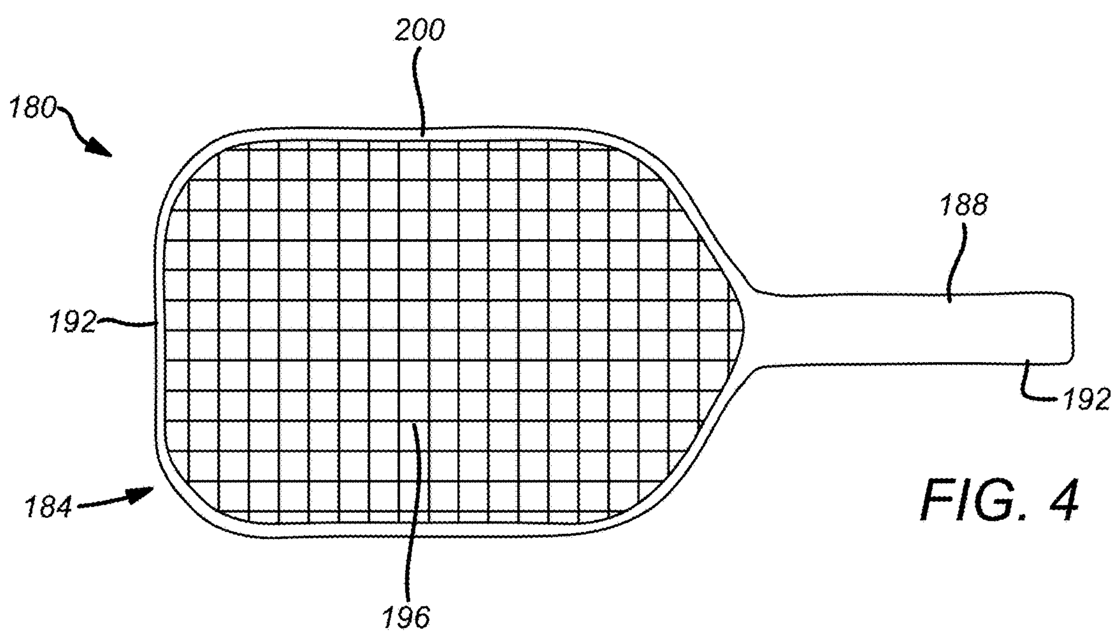




FIG. 4

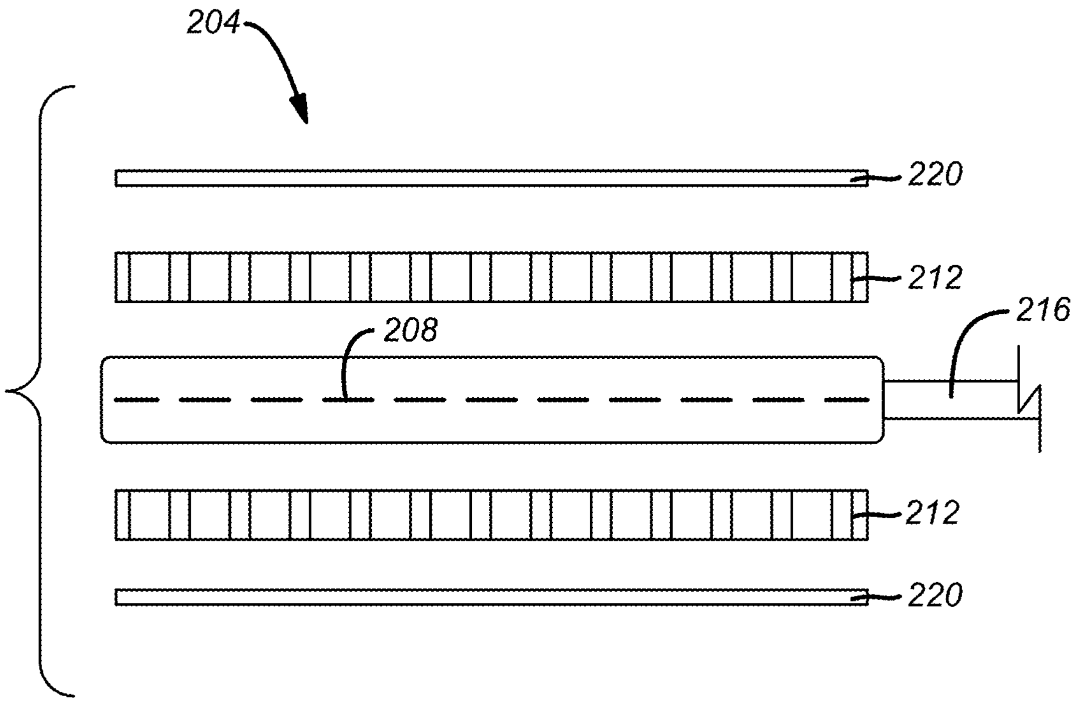



FIG. 5

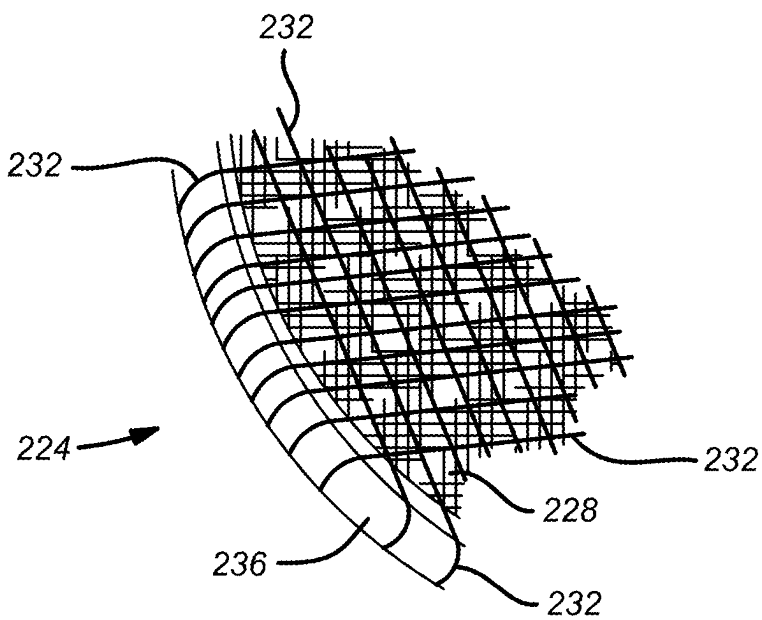



FIG. 6

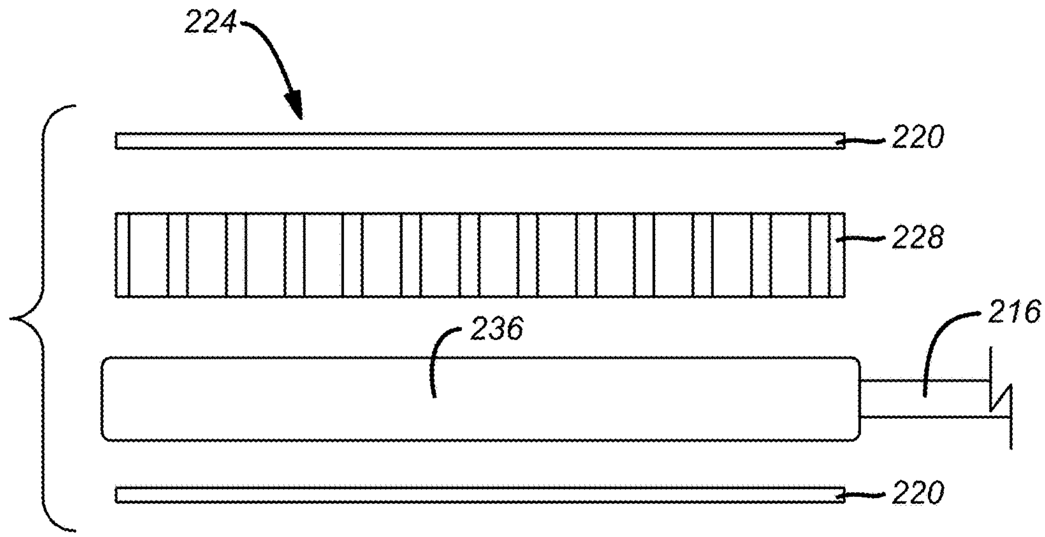



FIG. 7

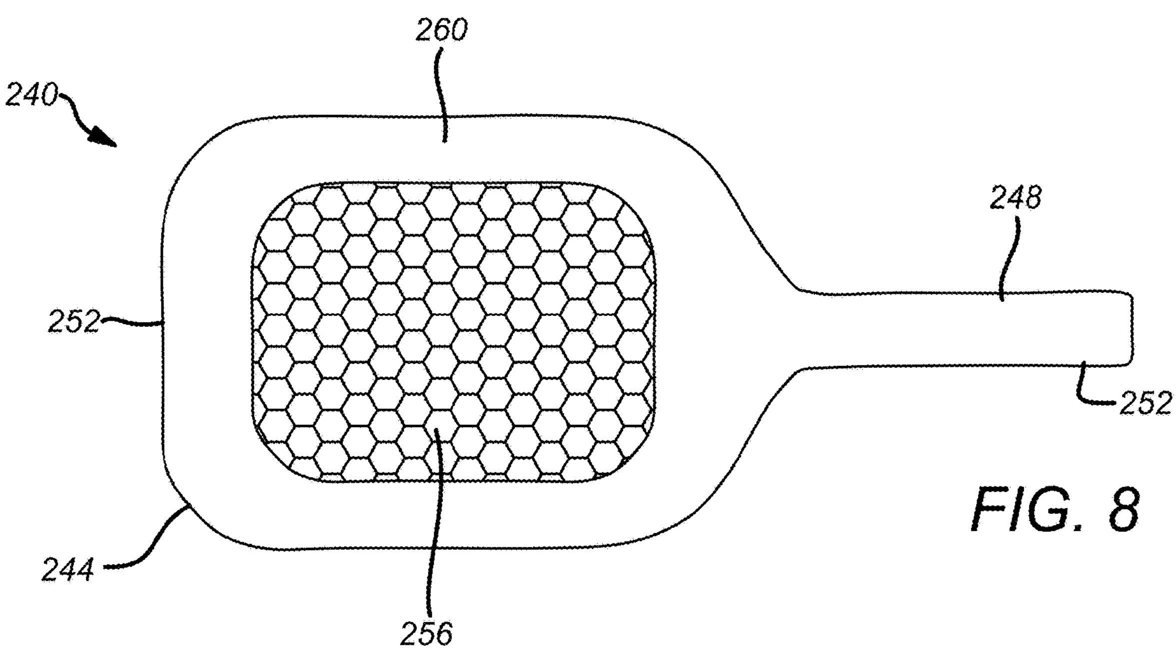
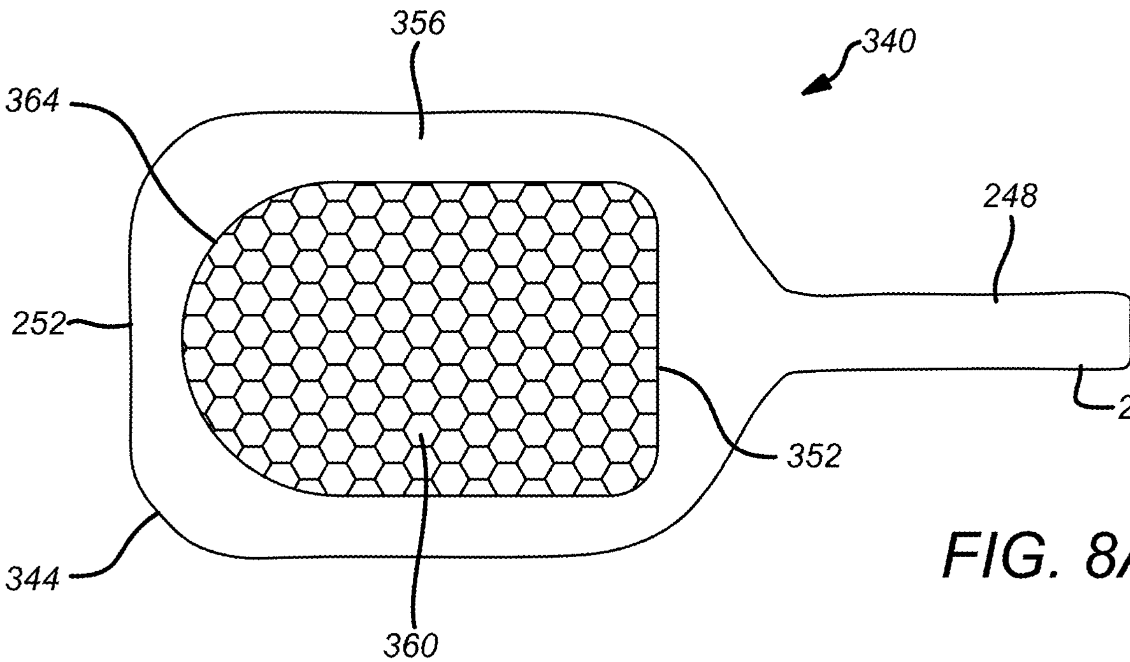



FIG. 8


356
340
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352
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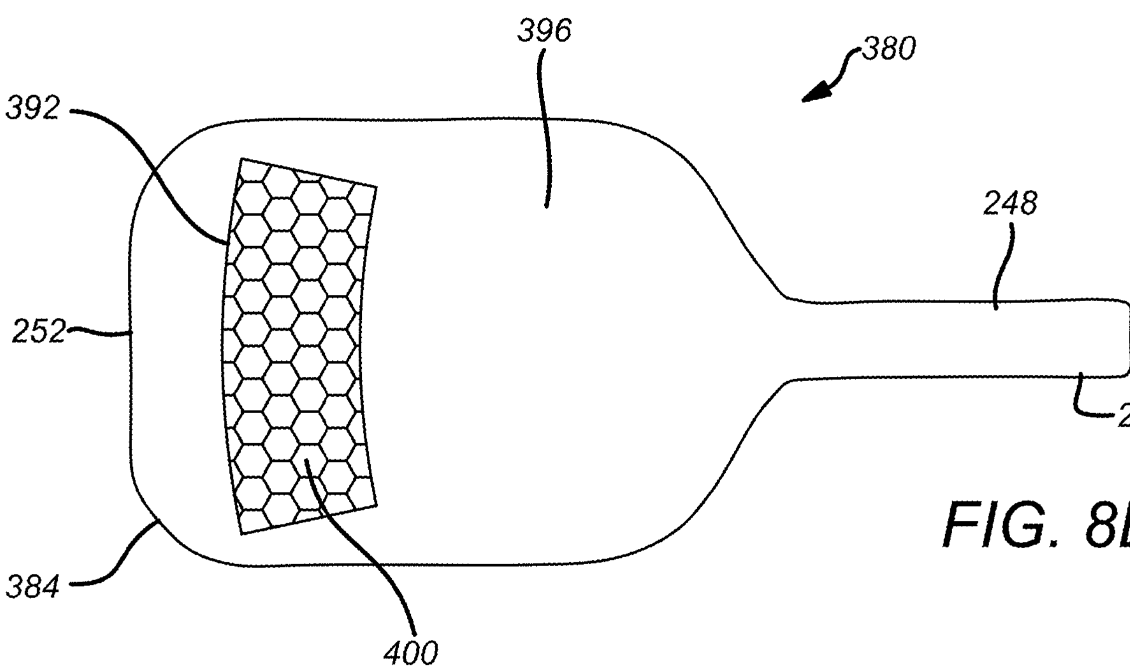
344
360

FIG. 8A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384
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FIG. 8B

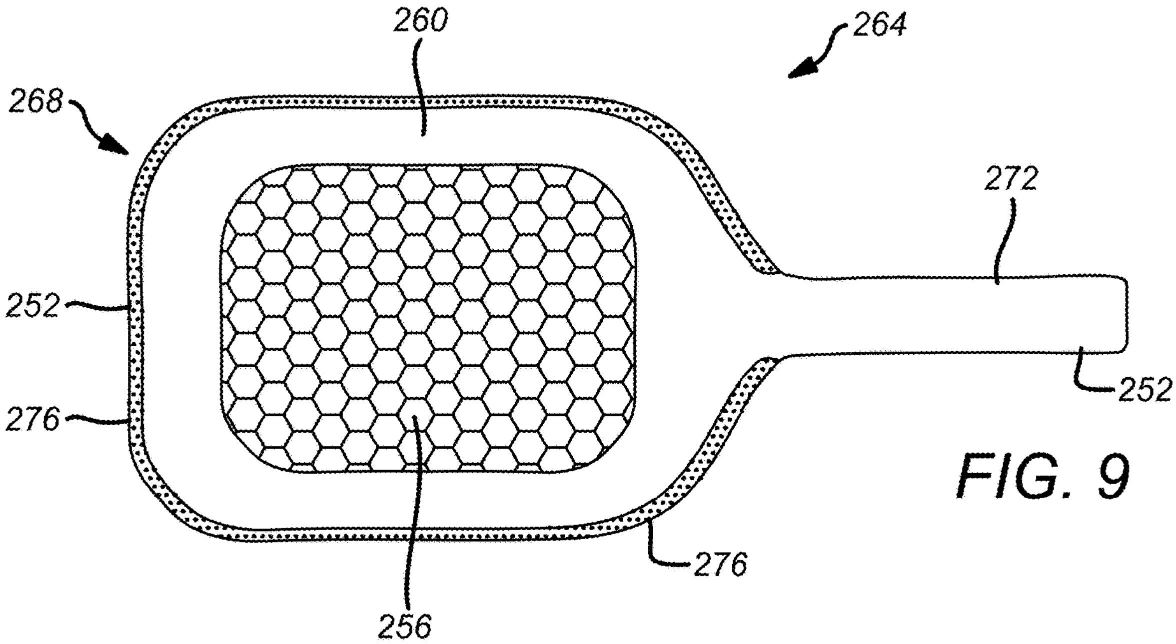
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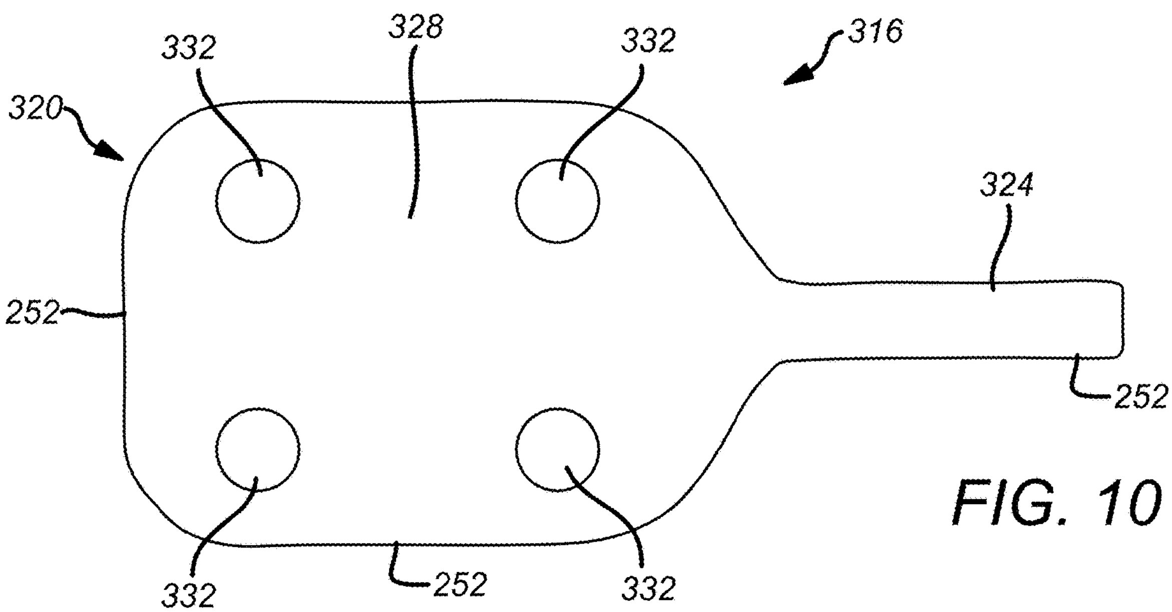
FIG. 9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252
332

FIG. 10

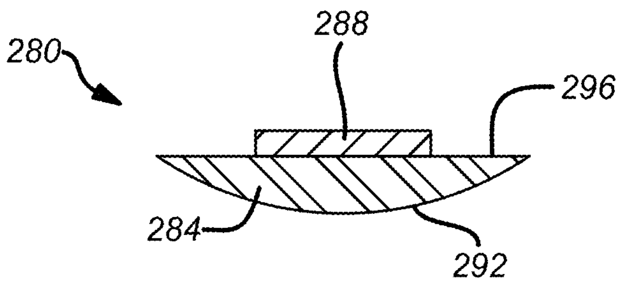



FIG. 11A

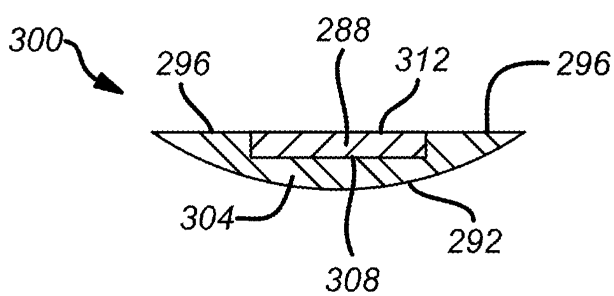



FIG. 11B

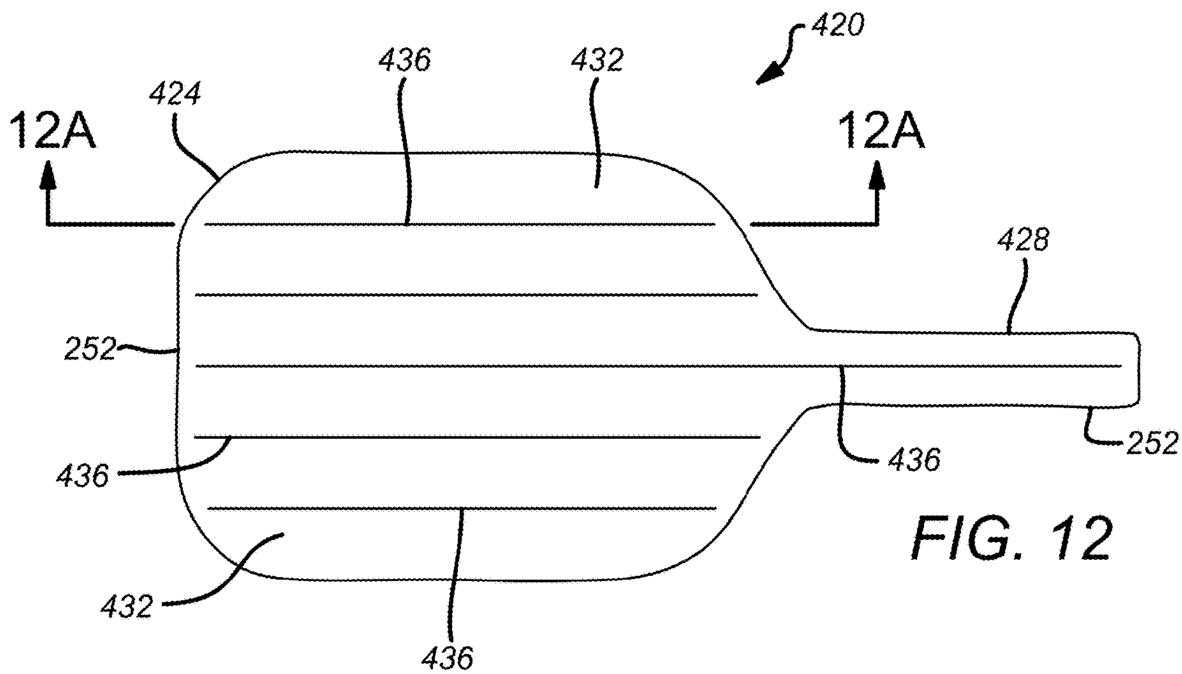



FIG. 12

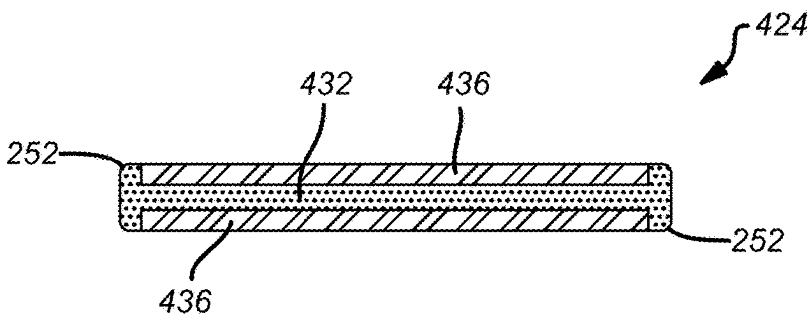
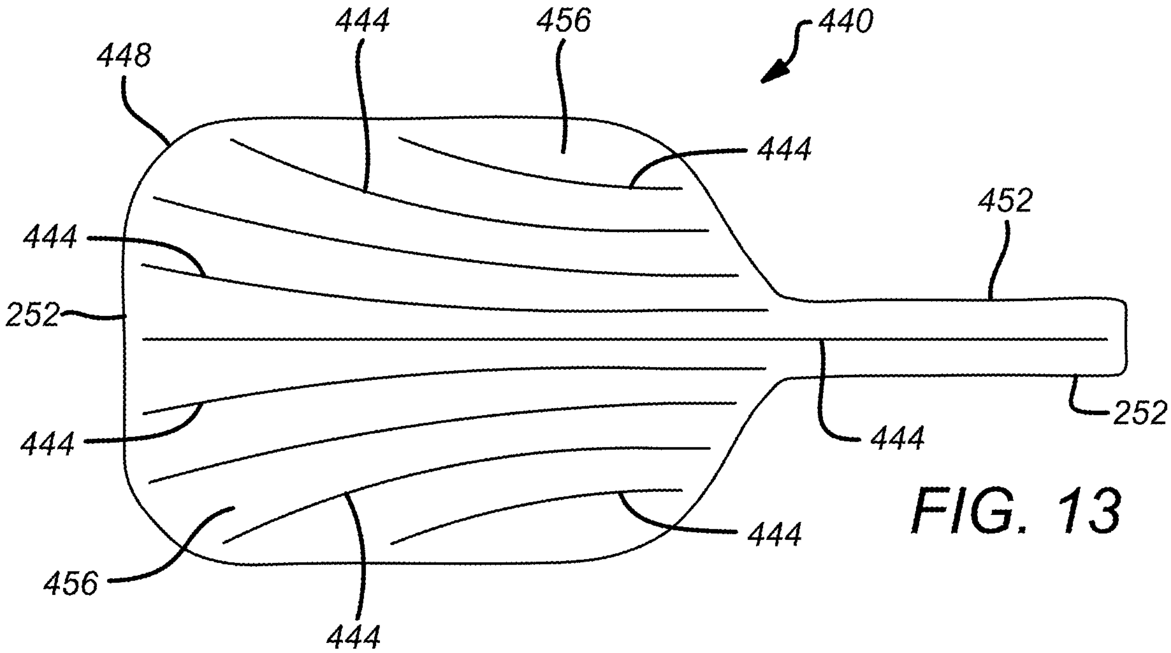



FIG. 12A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452
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252
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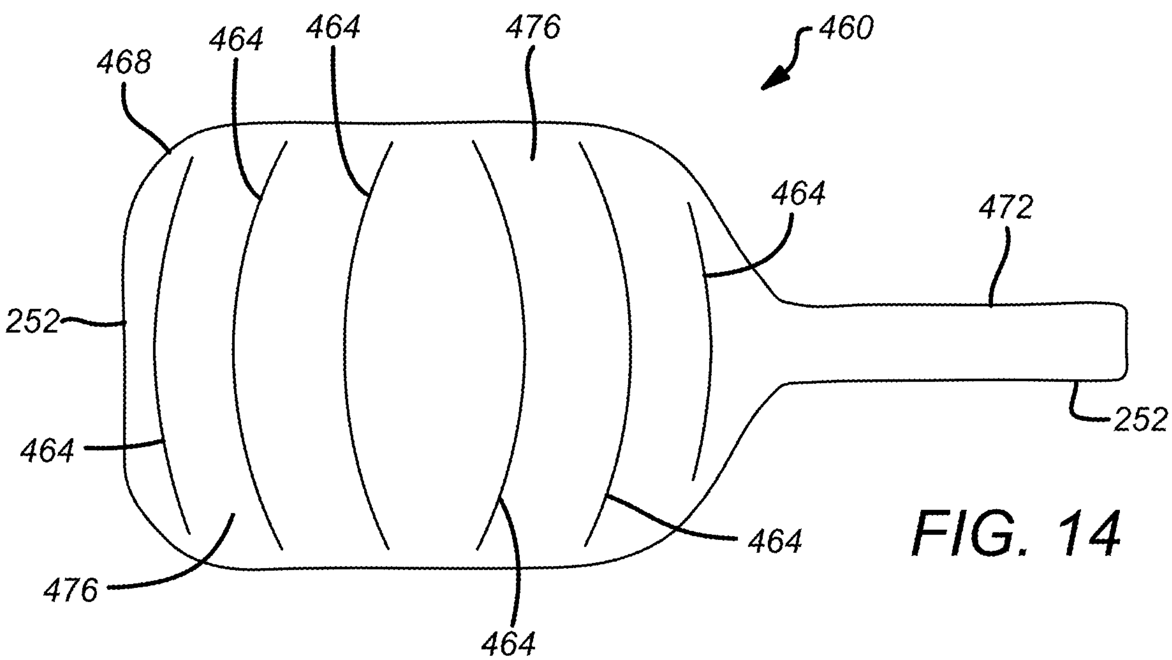
456
444
444

FIG. 13


464
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476
464

FIG. 14

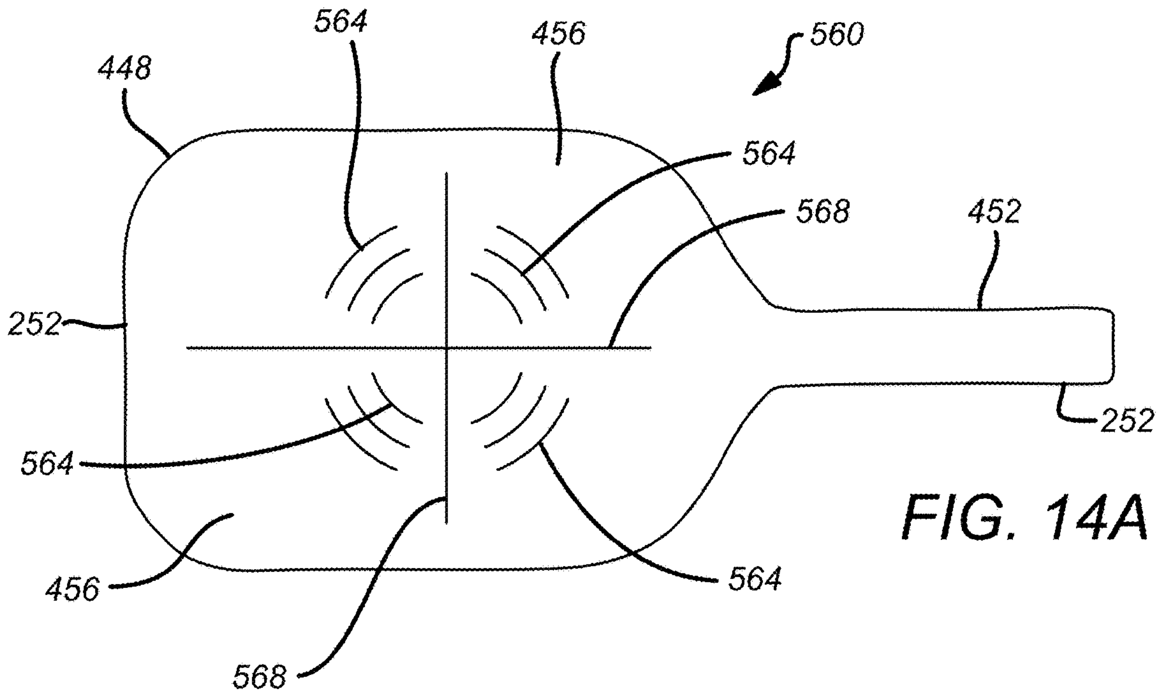


FIG. 14A

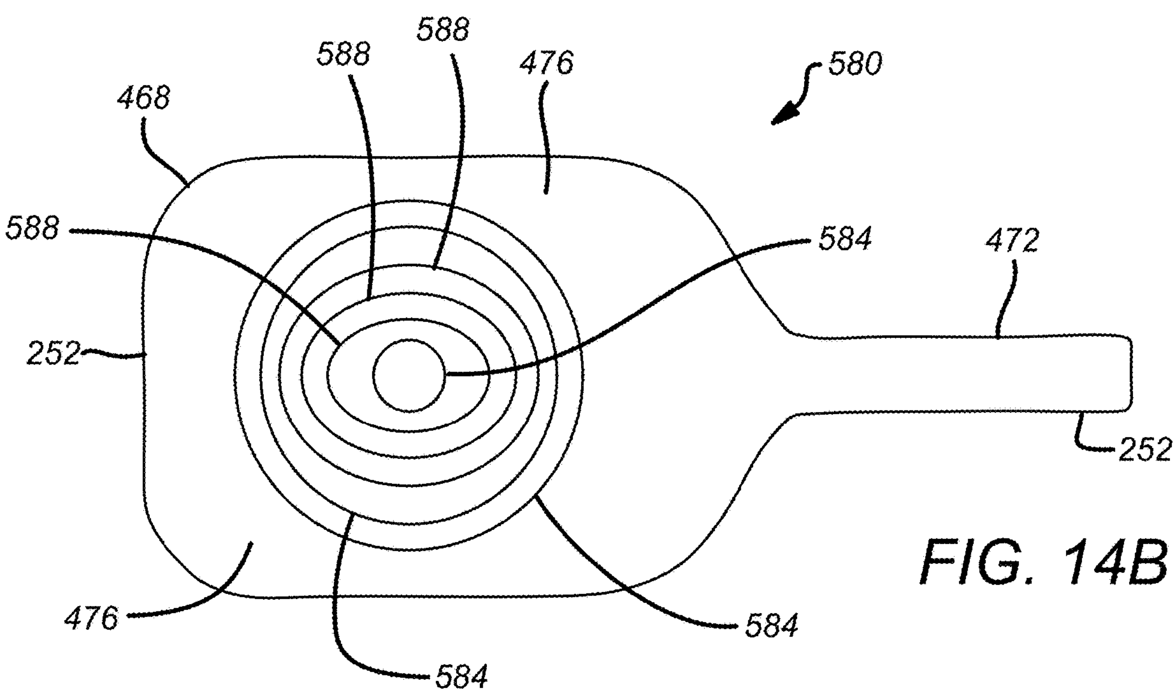


FIG. 14B

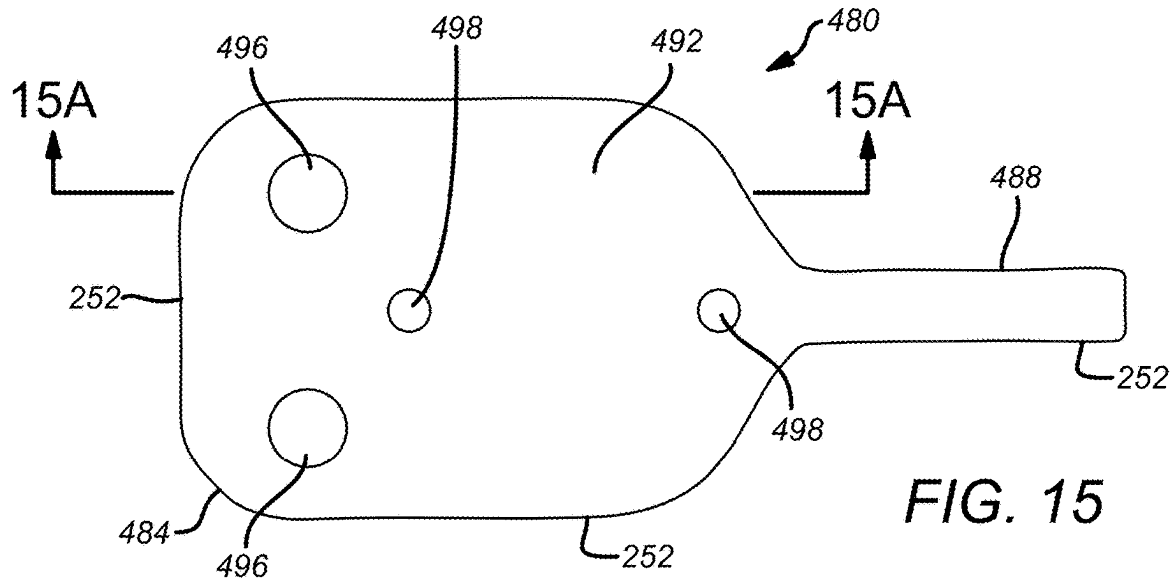


FIG. 15

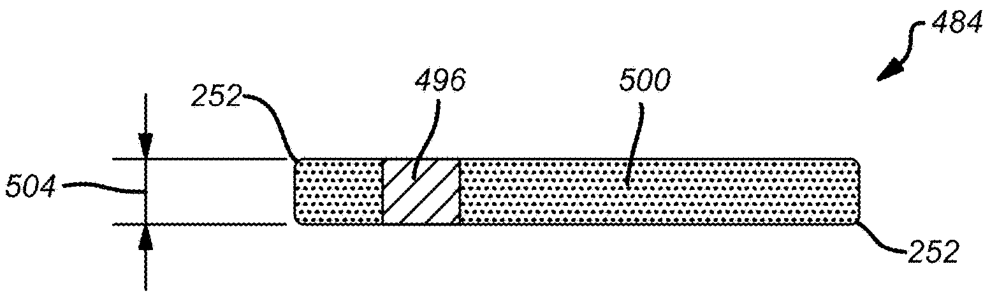


FIG. 15A

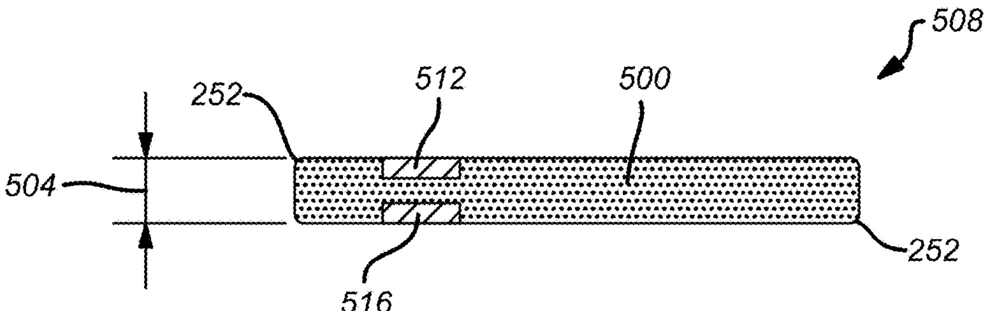


FIG. 15B

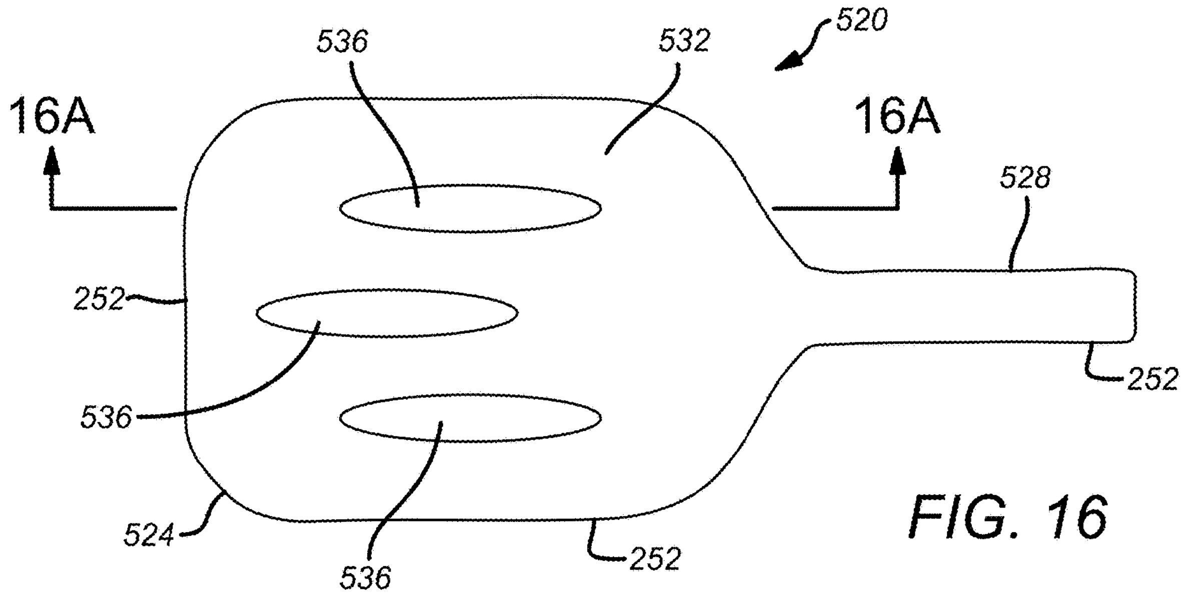


FIG. 16

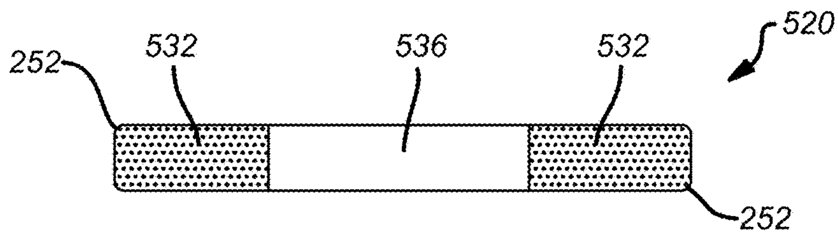


FIG. 16A

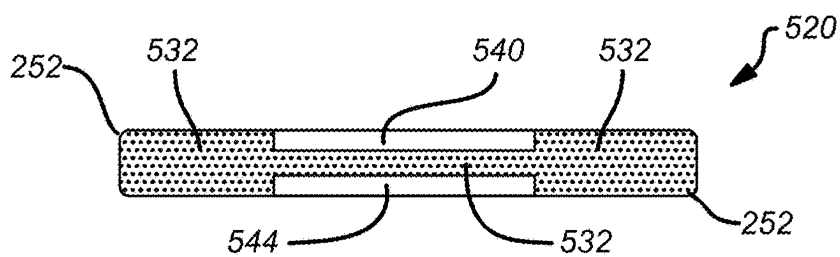


FIG. 16B

# SPORTS PADDLE

## PRIORITY

**[0001]** This application is a continuation-in-part of, and claims the benefit of, U.S. patent application, entitled "Sports Paddle," filed on May 8, 2024, and having application Ser. No. 18/658,701, which claims the benefit of, and priority to, U.S. patent application, entitled "Sports Paddle," filed on May 8, 2023, and having application Ser. No. 18/144,770, which claims the benefit of, and priority to, U.S. Provisional application, entitled "Sports Paddle," filed on May 9, 2022, and having application Ser. No. 63/339,855, and to U.S. Provisional application, entitled "Sports Paddle," filed on Oct. 15, 2022, having application Ser. No. 63/419,154, and to U.S. Provisional application, entitled "Core Strung Sports Paddle," filed on Feb. 5, 2025, having application Ser. No. 63/754,416, the entirety of each of said applications being incorporated herein by reference.

## FIELD

**[0002]** Embodiments of the present disclosure generally relate to paddles for sports. More specifically, embodiments of the disclosure relate to an apparatus and methods for a sports paddle for pickleball, tennis, smashball, paddle tennis, ping pong, or racquetball.

## BACKGROUND

**[0003]** Pickleball typically is played on a badminton-sized court with a tennis-like net, using a perforated plastic hollow ball. The ball is struck by players using pickleball paddles that resemble large table tennis paddles. Pickleball paddles generally are be made from wood, graphite, aluminum, carbon fiber, fiberglass, composite materials, and other suitable materials. They often have a polypropylene or foam core.

**[0004]** Pickleball paddles may be constructed by cutting or machining a large sheet of material into the shape of a paddle or by surrounding the core in a composite material and allowing it to cure in a fixed tool. As shown in FIG. **1**, a conventional paddle **100** generally includes a ball striking, planar portion **104** and a handle portion **108**. The planar portion **104** defines opposing flat surfaces. An outer edge guard **112** generally is provided around the planar portion **104** to protect core material comprising the planar portion **104**. The planar portion **104** extends downward into the handle portion **108**. A synthetic or polymeric grip **116** is wrapped around the handle portion **108** to provide stability and firmness for the player. A binding **120** secures the grip **116** to a neck **124** of the paddle **100**. An end cap **128** placed at an end of the handle **108** provides support and protection of the paddle **100**.

**[0005]** The opposing flat surfaces of the paddle **100** may be covered by a skin or protective layers. The skin is sometimes called a "laminate" or "face sheet." In embodiments where a core material is bonded to a skin material, the paddle **100** is referred to as a "composite panel" or "sandwich panel." The skin material may comprise thin, but stiff, skins while the core material may comprise a lightweight, but thick, core. The core material generally is low strength material, but its greater thickness provides the sandwich panel with high bending stiffness and low overall density.

**[0006]** The core material may comprise open-cell-structures and closed-cell-structured foams such as polyvinylchloride, polyurethane, polyethylene or polystyrene foams, balsa wood, syntactic foams, honeycombs, and the like. Open-cell and closed-cell metal foam can also be used as core materials.

**[0007]** Laminates comprising glass or carbon fiber-reinforced thermoplastics or thermoset polymers, such as unsaturated polyesters, epoxies, and the like, may be used as skin materials. Sheet metal may also be used as skin material.

**[0008]** A drawback to conventional paddles is that they tend to develop weak spots or soft spots, known as "dead spots." Dead spots typically form at locations where wear damage occurs, or where the adhesive fails, resulting in delamination of the skin material from the core material. Dead spots can also occur because edges of conventional paddles are held together only by a flexible rail guard that allows the paddle edges to flex during ball impacts.

**[0009]** Another drawback to conventional paddles comprising sandwich panels, such as the paddle **100**, arises due to edge treatments. Some manufacturers apply a U-channel or other type of molding, such as the edge guard **112**, to protect the edge of the planar portion **104** and enclose the periphery of the core material disposed between the skin materials. The edge guard **112** forms a lip **132** around the planar portion **104** that creates interference when striking the ball in this area of the paddle **100**. Further, some manufacturers finish the edge with a composite material. Unfortunately, the composite material renders the edge of the paddle **100** overly fragile, resulting in damage if the ground or a hard object is struck during play. Moreover, many methods of finishing the edge of the paddle **100** cause reliability issues, such as the edge treatment tending loosen and separate or fall off with age or stress. Additionally, in some instances a flexible rail guard may be used, but such rail guards often cause undesirable ball trajectories when struck by the ball during play.

**[0010]** There is a need, therefore, for a paddle that does not require an edge treatment, such as the edge guard **112** shown in FIG. **1**.

## SUMMARY

**[0011]** An apparatus and a method are provided for a sports paddle for pickleball, tennis, smashball, ping pong, or racquetball. The sports paddle comprises a planar portion defining opposing flat surfaces for striking a ball and a handle portion for grasping in a hand. A skin layer encloses the planar portion and the handle portion. Rounded edges are disposed around the planar portion and the handle portion. The planar portion may comprise an internal string core coupled with structured foam layers and surrounded by carbon fiber layers. The planar portion may comprise a core material surrounded by a rigid portion. The rigid portion may comprise 20-24 lb. foam while the core material is a honeycomb structured material. A synthetic or polymeric grip is wrapped around the handle portion to provide stability and firmness for a player. A binding secures the grip to a neck of the planar portion. An end cap placed at an end of the handle provides support and protection of the paddle.

**[0012]** In an exemplary embodiment, a sports paddle comprises: a planar portion having opposing flat surfaces for striking a ball; a core material having rounded edges; a handle portion for grasping in a hand; and a skin layer surrounding at least a majority of the planar portion.

[0013] In another exemplary embodiment, the planar portion includes a shaped core material and a rigid portion. In another exemplary embodiment, the shaped core material comprises a window-like structure that provides a planar portion that exhibits a relatively greater degree of rigidity.

[0014] In another exemplary embodiment, one or more weights are incorporated into the planar portion to increase the weight of the paddle. In another exemplary embodiment, any one or more of the one or more weights comprises a stabilizer that provides desired performance results. In another exemplary embodiment, any one or more of the multiple weights and the stabilizers comprise slugs of gel rubber for weighing the paddle and/or creating a dampening of vibration. In another exemplary embodiment, any one or more of the multiple weights and the stabilizers comprises a first weight and a second weight that are embedded on opposite sides of the core.

[0015] In another exemplary embodiment, the planar portion comprises multiple stringers that are embedded in the core material. In another exemplary embodiment, the multiple stringers comprise strips of wood or plastic that are glued into the core material. In another exemplary embodiment, the multiple stringers comprise rubber, a foam, or another suitable material that provides a flexible planar portion. In another exemplary embodiment, one or more of the multiple stringers comprises a curved stringer that is embedded into the planar portion.

[0016] In another exemplary embodiment, the planar portion comprises a rigid portion that includes a core material having one or more channels that are arranged in a pattern that desirably enhances the performance of the paddle. In another exemplary embodiment, any one or more of the one or more channels extends all the way through the core material. In another exemplary embodiment, any one or more of the one or more channels comprises a first channel and a second channel that are disposed on opposite sides of the planar portion, leaving a relatively thin portion of the core material disposed between the first channel and the second channel.

[0017] In another exemplary embodiment, the planar portion includes an internal string core surrounded by structured foam layers. In another exemplary embodiment, the string core is strung on a rigid portion comprising carbon fiber, aluminum, or another similar material capable of supporting the string core. In another exemplary embodiment, a rigid layer is disposed outside each structured foam layer.

[0018] In another exemplary embodiment, the planar portion includes a core material surrounded by an external string core that is strung around the exterior of a rigid portion. In another exemplary embodiment, the rigid portion comprises carbon fiber, aluminum, or another similar material. In another exemplary embodiment, the external string core and the rigid portion surround an interior volume that houses the core material.

[0019] These and other features of the concepts provided herein may be better understood with reference to the drawings, description, and appended claims.

## BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The drawings refer to embodiments of the present disclosure in which:

[0021] FIG. **1** illustrates a perspective view of an exemplary embodiment of a conventional paddle that includes an edge guard;

[0022] FIG. **2** illustrates a side view of an exemplary embodiment of a sports paddle, in accordance with the present disclosure;

[0023] FIG. **3** illustrates an edge view of an exemplary embodiment of a sports paddle that includes rounded edges, according to the present disclosure;

[0024] FIG. **4** illustrates a side view of an exemplary embodiment of a sports paddle that includes a core material in accordance with the present disclosure;

[0025] FIG. **5** illustrates an exploded side view of an exemplary embodiment of a sports paddle that includes an internal string core surrounded by structured foam layers in accordance with the present disclosure;

[0026] FIG. **6** illustrates a close-up view of an exemplary embodiment of a sports paddle that includes a core material surrounded by an external string core in accordance with the present disclosure;

[0027] FIG. **7** illustrates an exploded side view of an exemplary embodiment of a sports paddle that includes a core material surrounded by an external string core according to the present disclosure;

[0028] FIG. **8** illustrates a side view of an exemplary embodiment of a sports paddle that includes a core material in accordance with the present disclosure;

[0029] FIG. **8**A illustrates a side view of an exemplary embodiment of a sports paddle that includes a shaped core material and a rigid portion in accordance with the present disclosure;

[0030] FIG. **8**B illustrates a side view of an exemplary embodiment of a sports paddle that includes a shaped core material and a rigid portion according to the present disclosure;

[0031] FIG. **9** illustrates a side view of an exemplary embodiment of a sports paddle that includes a core material and a padded edge guard in accordance with the present disclosure;

[0032] FIG. **10** illustrates a side view of an exemplary embodiment of a sports paddle that includes multiple weights according to the present disclosure;

[0033] FIG. **11**A illustrates a cross-sectional view of an exemplary embodiment of an edge guard that may be coupled with an exterior of a periphery of the sports paddle;

[0034] FIG. **11**B illustrates a cross-sectional view of an exemplary embodiment of an edge guard that may be coupled with an exterior of a periphery of the sports paddle;

[0035] FIG. **12** illustrates a side view of an exemplary embodiment of a sports paddle that includes stringers of a rigid material according to the present disclosure;

[0036] FIG. **12**A illustrates a cross-sectional view of the sport paddle of FIG. **12**, taken along line **12**A-**12**A, in accordance with the present disclosure;

[0037] FIG. **13** illustrates a side view of an exemplary embodiment of a sports paddle that includes curved stringers of a rigid material according to the present disclosure;

[0038] FIG. **14** illustrates a side view of an exemplary embodiment of a sports paddle that includes curved stringers of a rigid material in accordance with the present disclosure;

[0039] FIG. **14**A illustrates a side view of an exemplary embodiment of a sports paddle that includes a combination of curved stringers and linear stringers of a rigid material according to the present disclosure;

[0040] FIG. **14**B illustrates a side view of an exemplary embodiment of a sports paddle that includes a combination of circular stringers and elliptical stringers of a rigid material accordance with the present disclosure;

**[0041]** FIG. **15** illustrates a side view of an exemplary embodiment of a sports paddle that includes multiple stabilizer slugs or balance slugs according to the present disclosure;

**[0042]** FIG. **15**A illustrates a cross-sectional view of the sport paddle of FIG. **15**, taken along line **15**A-**15**A, in accordance with the present disclosure;

**[0043]** FIG. **15**B illustrates a cross-sectional view of an alternative embodiment of the sport paddle of FIG. **15**, taken along line **15**A-**15**A, according to the present disclosure;

**[0044]** FIG. **16** illustrates a side view of an exemplary embodiment of a sports paddle that includes channels or holes according to the present disclosure;

**[0045]** FIG. **16**A illustrates a cross-sectional view of the sport paddle of FIG. **16**, taken along line **16**A-**16**A, in accordance with the present disclosure; and

**[0046]** FIG. **16**B illustrates a cross-sectional view of an alternative embodiment of the sport paddle of FIG. **16**, taken along line **16**A-**16**A, according to the present disclosure.

**[0047]** While the present disclosure is subject to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. The present disclosure should be understood to not be limited to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure.

## DETAILED DESCRIPTION

**[0048]** In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be apparent, however, to one of ordinary skill in the art that the paddle and methods disclosed herein may be practiced without these specific details. In other instances, specific numeric references such as "first core material," may be made. However, the specific numeric reference should not be interpreted as a literal sequential order but rather interpreted that the "first core material" is different than a "second core material." Thus, the specific details set forth are merely exemplary. The specific details may be varied from and still be contemplated to be within the spirit and scope of the present disclosure. The term "coupled" is defined as meaning connected either directly to the component or indirectly to the component through another component. Further, as used herein, the terms "about," "approximately," or "substantially" for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or collection of components to function for its intended purpose as described herein.

**[0049]** Pickleball paddles generally are be made from wood, graphite, aluminum, fiberglass, carbon fiber, composite materials, and other suitable materials. In many embodiments, a core material is bonded between sheets of a skin material. A drawback to conventional paddles is that they tend to develop dead spots at locations where wear damage occurs, or where the adhesive fails, resulting in delamination of the skin material from the core material. Another drawback arises due to edge treatments, such as an edge guard used to protect the edge of the planar portion and enclose the periphery of the core material disposed between the skin materials. The edge guard forms a lip around the planar portion that creates interference when striking the ball in this area of the paddle. Embodiments presented herein provide a paddle that does not require an edge guard, allowing the paddle to be made out of a continuous material that encloses the core material as well as any structures incorporated into the core.

**[0050]** FIG. **2** illustrates an exemplary embodiment of a paddle **140** that obviates any need for an edge treatment, such as the edge guard **112**, in accordance with the present disclosure. The paddle **140** includes a planar portion **144** and a handle portion **148**. The planar portion **144** defines opposing flat surfaces for striking a pickleball. The planar portion **144** comprises a core material, as described herein. The planar portion **144** extends into the handle portion **148**. A synthetic or polymeric grip **152** is wrapped around the handle portion **148** to provide stability and firmness for the player. In some embodiments, the grip **152** may be wrapped around hardwood portions (not shown) that may be coupled with the handle portion **148**. In some embodiments, the grip **152** may be wrapped around portions comprising EVA foam (not shown), or another suitable material, that may be coupled with the handle portion **148**. It is contemplated that a wide variety of materials may comprise the portions coupled with the handle portion **148**, without limitation. A binding **156** secures the grip **152** to a neck **160** of the paddle **140**. An end cap **164** placed at an end of the handle **148** provides support and protection of the paddle **140**.

**[0051]** As shown in FIG. **2**, the core material comprising paddle **140** is wrapped within a skin layer **168**. Unlike conventional paddles that comprise a sandwich panel structure, the skin layer **168** may surround an entirety of the core material. As shown in FIG. **3**, enclosing the core material within the skin layer **168** enables incorporating rounded edges **172** into the planar portion **144** and the handle portion **148** without requiring an edge treatment, such as the edge guard **112** shown in FIG. **1**. Once the skin layer **168** is applied to the planar portion **144** and the handle portion **148**, the grip **152**, the binding **156**, and the end cap **164** may be installed onto the paddle **140** as shown in FIG. **2**.

**[0052]** The skin layer **168** may comprise fiberglass, graphite, carbon fiber, or other suitable material that offers a practitioner a sense of control, feel, or power during play. In some embodiments, the opposing sides of the planar portion **144** may be covered by skin layers **168** comprising different materials, thereby forming a "hybrid" paddle that offers differing performance on the opposing sides. For example, one side may offer greater power while the other side offers greater precision. Further, in some embodiments, the skin layers **168** may be laminates comprising glass or carbon fiber-reinforced thermoplastics or thermoset polymers, such as unsaturated polyesters, epoxies, and the like, as well as sheet metal, without limitation. Further, in some embodiments, the skin layers **168** may be configured to provide specific performance properties, such as, for example, a consistent traction. In some embodiments, the skin layers **168** may include permanent or removeable film portions that are configured to provide desired performance properties, such as a consistent traction, without limitation. In some embodiments, adhesive stickers comprising carbon fiber sheets may be configured to provide consistent or enhanced traction. It is contemplated that such adhesive stickers may be applied to and removed from the planar portion **144** of the paddle **140** as desired.

[0053] In some embodiments, the skin layer **168** may be configured to provide noise suppression properties. As such, in some embodiments, the skin layer **168** may include adhesive stickers comprising a synthetic rubber compound, such as by way of non-limiting example, neoprene or ethylene propylene diene monomer (EPDM), and the like. It is contemplated that, in some embodiments, the noise suppressing stickers can be applied to and removed from the planar portion **144** of the paddle **140**, as desired. In some embodiments, the skin layer **168** may comprise fixed foam surfaces that cover the planar portion **144** of the paddle **140**. Further, in some embodiments, the skin layer **168** may comprise a removeable skin that is configured to be pulled onto the planar portion **144** and then closed onto the paddle **140**. It is contemplated that the removeable skin includes a zipper or similar device for enclosing the planar portion **144**. In some embodiments, the removeable skin may include foam portions configured to dampen the sound of striking a ball with the paddle **140**. As will be appreciated, applying noise damping properties to the paddle **140** is particularly useful for playing pickleball or other similar games near residences.

[0054] The core material may comprise open-cell-structures and closed-cell-structured foams such as polyvinylchloride, polyurethane, polyethylene or polystyrene foams, balsa wood, syntactic foams, honeycombs, and the like. Open-cell and closed-cell metal foam can also be used as core materials. In one embodiment, the core material comprises a complete polypropylene honeycomb core that is wrapped in 4 layers of pre-preg carbon fiber. The carbon fiber is adhered to the honeycomb and baked under compression to form a "boned surface." In one embodiment, the core material comprises between 20-lb. and 24-lb. foam. Other suitable materials will be apparent to those skilled in the art.

[0055] Moreover, in some embodiments, the polypropylene honeycomb is dipped into a bath of a liquid material that remains flexible in a dried state. The liquid material may comprise any one or more of rubber, Urethanes, as well as exotic materials such as liquid diamond dust, liquid carbon fiber, liquid Kevlar, liquid gold dust, liquid ruby dust, and the like. Once the liquid material is dried onto the honeycomb core, the core is wrapped in 4 layers of pre-preg carbon fiber. The carbon fiber is adhered to the honeycomb and baked under compression to form a boned surface, as mentioned above. It is contemplated that the dried liquid material can be used to change the properties of the paddle **140**, such as change weight, construction strength, bounce, and the like. Further, it is contemplated that dipping the honeycomb core seals off the honeycomb chamber walls, preventing small pieces from coming off causing rattles.

[0056] It is contemplated that any of various structures may be incorporated into the planar portion before being wrapped in the skin layers **168**, as described herein. For example, FIG. **4** illustrates an exemplary embodiment of a paddle **180** that includes a structured planar portion **184** and a handle portion **188** in accordance with the present disclosure. The planar portion **184** and the handle portion **188** have rounded edges **192**, as described herein. The planar portion **184** comprises a core **196** surrounded by a rigid portion **200**. The core **196** and the rigid portion **200** may comprise any of various materials that are found to provide a product that can be more efficiently produced, that is more aesthetically pleasing, that provides greater consistency of function, and is more reliable than products produced using other technologies. In some embodiments, the rigid portion **200** comprises a carbon fiber or aluminum frame while the core **196** comprises an internal string core that is sandwiched between polypropylene or foam layers.

[0057] FIG. **5** illustrates an exploded side view of an exemplary embodiment of a sports paddle **204** that includes an internal string core **208** surrounded by structured foam layers **212** in accordance with the present disclosure. The string core **208** is strung, and dynamically tensioned, on a rigid portion **216**. Like the rigid portion **200** mentioned above, the rigid portion **216** comprises carbon fiber, aluminum, or another similar material capable of supporting the string core **208**. A structured foam layer **212** is positioned above and below the string core **208**. The structured foam layers **212** can comprise any of polyvinylchloride, polyurethane, polyethylene or polystyrene foams, balsa wood, syntactic foams, honeycombs, and the like, without limitation. As shown in FIG. **5**, a rigid layer **220** is disposed outside each structured foam layer **212**. The rigid layers **220** can comprise multilayer carbon fiber, fiberglass, or another similar material, without limitation. It is contemplated that, in some embodiments, the paddle **204** can be wrapped within a skin layer **168**, as described herein.

[0058] FIG. **6** illustrates a close-up view of an exemplary embodiment of a sports paddle **224** that includes a core material **228** surrounded by an external string core **232** in accordance with the present disclosure. As shown in FIG. **6**, the string core **232** is strung around the exterior of a rigid portion **236**. The rigid portion **236** can comprise carbon fiber or another similar material. In some embodiments, the rigid portion **236** comprises aluminum or a suitable composite material. It is contemplated that the string core **232** can be strung on either one or both sides of the rigid portion **236**. In embodiments wherein the string core **232** is strung on both sides of the rigid portion **236**, the string core **232** and the rigid portion **236** surround an interior volume that can house the core material **228**.

[0059] FIG. **7** illustrates an exploded side view of an exemplary embodiment of a sports paddle **224** that includes a core material **228** surrounded by an external string core **232** that is strung on a rigid portion **236** according to the present disclosure. As mentioned above, the rigid portion **236** can comprise, in some embodiments, aluminum or a suitable composite material. The string core **232** can be strung on one or both sides of the rigid portion **236**. As shown in FIG. **7**, the core material **228** is disposed within the rigid portion **236**. In embodiments wherein the string core **232** is strung on both sides of the rigid portion **236**, the string core **232** and the rigid portion **236** surround an interior volume that houses the core material **228**. The core material **228** can comprise any of polyvinylchloride, polyurethane, polyethylene or polystyrene foams, balsa wood, syntactic foams, honeycombs, and the like, without limitation. As further shown in FIG. **7**, a rigid layer **220** is disposed outside the core material **228** and the string core **232**. The rigid layers **220** can comprise multilayer carbon fiber, fiberglass, or another similar material, without limitation. It is contemplated that, in some embodiments, the paddle **224** can be wrapped within a skin layer **168**, as described herein.

[0060] As mentioned hereinabove, it is contemplated that any of various structures may be incorporated into the planar portion before being wrapped in the skin layers **168**, as described herein. For example, FIG. **8** illustrates an exemplary embodiment of a paddle **240** that includes a structured planar portion **244** and a handle portion **248** in accordance with the present disclosure. The planar portion **244** and the handle portion **248** have rounded edges **252**, as described herein. The planar portion **244** comprises a core **256** surrounded by a rigid portion **260**. The core **256** and the rigid portion **260** may comprise any of various materials that are found to provide a product that can be more efficiently produced, that is more aesthetically pleasing, that provides greater consistency of function, and is more reliable than products produced using other technologies. In one exemplary embodiment, the rigid portion **260** comprises between 20-lb. and 24-lb. foam, while the core **256** comprises a honeycomb structured material.

**[0061]** It should be understood that the core **256** can be implemented with various different shapes in the planar portion **244** to achieve performance objectives, such as power, touch, control, and the like. For example, FIG. **8**A illustrates an exemplary embodiment of a sports paddle **340** that includes a shaped core material and a rigid portion in accordance with the present disclosure. The paddle **340** is substantially similar to the paddle **240**, shown in FIG. **8**, with the exception that the paddle **340** comprises a planar portion **344** that includes a core shape **352** that is surrounded by a rigid portion **356**. The core shape **352** includes a core material **360** that can be strategically positioned in the planar portion **344** to provide the performance objectives mentioned above. As shown, for instance, the core shape **352** includes a rounded portion **364** that advantageously affects the performance of the paddle **340**. In some embodiments, the rigid portion **356** comprises between 20-lb. and 24-lb. foam, while the core material **360** comprises a honeycomb structured material.

**[0062]** FIG. **8**B illustrates another exemplary embodiment of a sports paddle **380** that includes a shaped core material and a rigid portion according to the present disclosure. The paddle **380** is substantially similar to the paddle **340**, shown in FIG. **8**A, with the exception that the paddle **380** comprises a planar portion **384** that includes a core shape **392** that is surrounded by a rigid portion **396**. The core shape **392** includes a core material **400** that can be strategically positioned in the planar portion **384** to provide the performance objectives mentioned with respect to FIG. **8**A. For example, the core shape **392** comprises a window-like structure that occupies relatively little space on the planar portion **384**, thus providing a planar portion **384** that exhibits a greater degree of rigidity than the paddle **340** shown in FIG. **8**A. Further, the rigid portion **396** may comprise between 20-lb. and 24-lb. foam, while the core material **400** comprises a honeycomb structured material.

**[0063]** FIG. **9** illustrates an exemplary embodiment of a paddle **264** that includes a structured planar portion **268** and a handle portion **272** in accordance with the present disclosure. The paddle **264** is substantially similar to the paddle **240** shown in FIG. **8**, with the exception that the paddle **264** of FIG. **9** includes a planar portion **268**, a handle portion **272**, and a padded edge guard **276**. The planar portion **268** and the handle portion **272** have rounded edges **252**, as described herein. The planar portion **268** comprises a core **256** surrounded by a rigid portion **260**, as described with respect to FIG. **8**. The core **256** and the rigid portion **260** may comprise any of various materials that are found to provide a product that can be more efficiently produced, that is more aesthetically pleasing, that provides greater consistency of function, and is more reliable than products produced using other technologies. It is contemplated that the padded edge guard **276** may be bonded to a periphery of the rigid portion **260** before the skin layer **168** is applied to the planar portion **268**. As such, the padded edge guard **276** can provide protection to the edge of the paddle **264** without any need for the bulky edge guard **112** discussed in connection with FIG. **1**.

**[0064]** Moreover, it is contemplated that an edge guard may be bonded to the periphery of the rigid portion **260** after the skin layer **168** is applied to the planar portion **268**. FIG. **11**A illustrates an exemplary embodiment of an edge guard **280** that may be coupled with an exterior of the periphery of the rigid portion **260**. The edge guard **280** is configured to provide protection to the portions of the skin layer **168** that are disposed on the rounded edges **252**. As shown in FIG. **11**A, the edge guard **280** comprises a polymer portion **284** and a mass strip portion **288**. The polymer portion **284** may be made of polyurethane or any other material that is suitable for protecting the rounded edges of the paddle **264**. In the embodiment of FIG. **11**A, the polymer portion **284** includes a rounded surface **292** that may be configured to smoothly join with the rounded edges **252** to produce a low-profile edge rail for the paddle **264**. The mass strip **288** preferably comprises a thin strip of metal, such as lead, that may be used to increase the weight of the paddle **264**. As shown in FIG. **11**A, the mass strip **288** may be coupled with a flat surface **296** of the polymer portion **284**. The mass strip **288** may be adhered to the flat surface **296** by way of any of various suitable adhesives. Further, the mass strip **288** and exposed portions of the flat surface **296** may be adhered to the periphery of the paddle **264** by way of any of various suitable adhesives, without limitation.

**[0065]** FIG. **11**B illustrates an exemplary embodiment of an edge guard **300** that may be coupled with an exterior of the periphery of the rigid portion **260**. The edge guard **300** is substantially similar to the edge guard **280** shown in FIG. **11**A, with the exception that the edge guard **300** comprises a polymer portion **304** that includes a recess **308** for receiving the mass strip **288**. The mass strip **288** may be adhered within the recess **308** by way of any suitable adhesive. As shown in FIG. **11**B, the recess **308** is configured to receive the mass strip **288** such that the flat surface **296** and a back surface **312** of the mass strip **288** form a substantially uniform surface suitable for being adhered to the periphery of the paddle **264**. The surfaces **296**, **312** may be adhered to the periphery of the paddle **264** by way of any suitable adhesive, without limitation. It is contemplated that disposing the mass strip **288** within the recess **308** will allow the rounded surface **292** of the edge guard **300** to smoothly join with the rounded edges **252** to produce a low-profile edge rail for the paddle **264**.

**[0066]** With reference, again, to FIGS. **8**-**9**, it should be borne in mind that the planar portion **268** is not limited to honeycomb structures. For example, in some embodiments, the planar portion **268** may comprise a solid portion of rigid foam, such as the abovementioned 20-24 lb. foam. Further, in some embodiments, the planar portion **268** may include a core **256** comprising a different type or density of foam than the foam comprising the rigid portion **260**. In some embodiments, the core **256** may comprise a number of holes advantageously positioned on the rigid portion **260** to reduce the weight of the paddle **264**.

**[0067]** In some embodiments, one or more weights may be incorporated into the planar portion to increase the weight of the paddle. For example, FIG. **10** illustrates an exemplary embodiment of a paddle **316** that includes multiple weights according to the present disclosure. The paddle **316** includes a planar portion **320** that extends to a handle portion **324**. Rounded edges **252** are disposed along all exterior edges of the planar portion **320** and the handle portion **324**. As shown in FIG. **10**, the planar portion **320** comprises a rigid portion **328** that includes multiple weights **332**. Although four weights **332** are shown, it is contemplated that the planar portion **320** may include any number of weights **332**. Further, any size and/or mass of the weights **332** may be disposed in any of various locations of the planar portion **320**, without limitation.

**[0068]** FIGS. **12** and **12**A illustrate an exemplary embodiment of a paddle **420** that includes a structured planar portion **424** and a handle portion **428** in accordance with the present disclosure. The planar portion **424** and the handle portion **428** have rounded edges **252**, as described herein. The planar portion **424** comprises multiple stringers **432** that are embedded in a core **436**. The core **436** and the stringers **432** may comprise any of various materials that are found to provide a product that can be more efficiently produced, that is more aesthetically pleasing, that provides greater consistency of function, and is more reliable than products produced using other technologies. As shown in FIG. **12**A, in some embodiments, the stringers **432** may comprise strips of wood or plastic that are glued into honeycomb or foam comprising the core **436**. It is contemplated that the rigidity of the stringers **432** can be used to provide a stronger planar portion **424**. More specifically, the stringers **432** may be used to enhance strength and core stability, facilitate customizable weight distributions, depending on stringer placement or material, improve rebound due to higher durometer materials, and boost durability by reinforcing weaker areas of the core **436**. In some embodiments, however, the stringers **432** may comprise rubber, a foam, or other suitable material that can provide a flexible planar portion **424**.

**[0069]** It should be borne in mind that the paddle **420** is not limited to the number of stringers **432** shown in FIG. **12**. Rather, any number of stringers **432** can be incorporated into the core **436** to provide a paddle **420** that exhibits the desired performance. For example, in some embodiments, the paddle **420** can include a single stringer **432** that extends along the midline of the paddle **420**. In some embodiments, the paddle **420** can include the single stringer **432** along the midline of the paddle **420** and two or more shorter stringers **432** that extend across the planar portion **424** parallel to the single stringer **432**.

**[0070]** Moreover, it is contemplated that, in some embodiments, the stringers **432** can be embedded into the planar portion **424** in different patterns that are found to provide desired performance results, such as enhanced power and control, improved balance and feel, and extended product life. FIG. **13**, for example, illustrates an exemplary embodiment of a paddle **440** that includes curved stringers **444** embedded into a planar portion **448** according to the present disclosure. Similar to the paddle **420**, shown in FIG. **12**, the paddle **440** includes rounded edges **252** disposed around the planar portion **448** and a handle portion **452**. The planar portion **448** comprises multiple curved stringers **444** that are embedded in a core **456**. As described with respect to FIG. **12**A, in some embodiments, the curved stringers **444** may comprise strips of wood or plastic that are glued into honeycomb or foam comprising the core **456**.

**[0071]** FIG. **14** illustrates an exemplary embodiment of a paddle **460** that includes curved stringers **464** embedded into a planar portion **468** accordance with the present disclosure. Similar to the paddle **440**, shown in FIG. **13**, the paddle **460** includes rounded edges **252** disposed around the planar portion **468** and a handle portion **472**. The planar portion **468** comprises multiple curved stringers **464** that are embedded in a core **476**. It is contemplated that the curved stringers **464** may comprise strips of wood or plastic that are glued into honeycomb or foam comprising the core **476**. As will be appreciated, the curved stringers **464** comprise a different orientation than the curved stringers **444** shown in FIG. **13**. It is contemplated that the curved stringers **464**, **444** can be arranged in any of various orientations and curvatures that are found to provide desired performance characteristics, without limitation.

**[0072]** In some embodiments, curved stringers and linear stringers can be combined to form paddles exhibiting desired performance characteristics, such as enhanced power and control, improved balance and feel, and extended product life, as mentioned above. For example, FIG. **14**A illustrates an exemplary embodiment of a paddle **560** that includes a combination of curved stringers **564** and linear stringers **568** that are embedded into a planar portion **448** according to the present disclosure. Similar to the paddle **440**, shown in FIG. **13**, the paddle **560** includes rounded edges **252** disposed around the planar portion **448** and a handle portion **452**. The planar portion **448**, shown in FIG. **14**A, comprises multiple curved stringers **564** and linear stringers **568** arranged to resemble a reticle that is embedded in a core **456**. The curved and linear stringers **564**, **568** may comprise strips of wood or plastic that are glued into honeycomb or foam comprising the core **456**.

**[0073]** FIG. **14**B illustrates an exemplary embodiment of a paddle **580** that includes a combination of circular stringers **584** and elliptical stringers **588** that are embedded into a planar portion **468** accordance with the present disclosure. Similar to the paddle **460**, shown in FIG. **14**, the paddle **580** includes rounded edges **252** disposed around the planar portion **468** and a handle portion **472**. In the embodiment of FIG. **14**B, the planar portion **468** comprises multiple circular stringers **584** that are concentric with multiple elliptical stringers **588** that are all embedded in a core **476**. As stated hereinabove, the circular and elliptical stringers **584**, **588** may comprise strips of wood or plastic that are glued into honeycomb or foam comprising the core **476**. In general, it should be appreciated that the stringers can be implemented in a wide variety of shapes and can be arranged in the core **476** with any of various orientations and curvatures that are found to provide desired performance characteristics, without limitation.

**[0074]** FIG. **15** illustrates an exemplary embodiment of a paddle **480** that includes multiple stabilizer slugs or balance slugs according to the present disclosure. The paddle **480** includes a planar portion **484** that extends to a handle portion **488**. Rounded edges **252** are disposed along all exterior edges of the planar portion **484** and the handle portion **488**. The planar portion **484** comprises a rigid portion **492** that includes multiple weights **496** and multiple stabilizers **498**. As will be appreciated, in some embodiments, the stabilizers **498** can comprise differently sized and/or shaped weights than the weights **496**. In some embodiments, one or more of the weights **496** and the stabilizers **498** may comprise different materials. In some embodiments, the weights **496** and the stabilizers **498** comprise slugs of gel rubber or a heavier material to weigh the paddle **480** or create dampening of vibration or both. Although two weights **496** and two stabilizers **498** are shown, it is contemplated that the planar portion **484** can include any number or combination of weights **496** and stabilizers **498**. Further, any size and/or mass of the weights **496** and the stabilizers **498** can be disposed in any of various locations of the planar portion **484**, without limitation.

**[0075]** It is contemplated that, in some embodiments, the weights **496** and/or the stabilizers **498** can extend partially or completely through the thickness of the planar portion **484**. For example, in an embodiment shown in FIG. **15**A, the weights **496** extend through a core **500** comprising the planar portion **484** and have a thickness **504** that is equal to the thickness of the planar portion **484**. Thus, once the paddle **480** is wrapped in skin layers **168**, as described herein, the weights **496** and the planar portion **484** provide a flat paddle surface, as shown in FIG. **3**. Similarly, it is contemplated that any one or more of the stabilizers **498** can be configured to extend through the core **500** as shown and discussed with respect to the weights **496**.

**[0076]** Moreover, the weights **496** and/or the stabilizers **498** need not extend all the way through the core **500**. For example, in an embodiment of the planar portion **508**, shown FIG. **15**B, a first weight **512** and a second weight **516** can be embedded on opposite sides of the core **500**, such that the weights **512**, **516** and the planar portion **508** provide a flat paddle surface once the paddle **480** is wrapped in skin layers **168**. Further, although the first and second weights **512**, **516** are shown in FIG. **15**B as having the same size, in some embodiments, the first and second weights **512**, **516** can have different shapes and sizes, as desired, without limitation.

**[0077]** FIG. **16** illustrates an exemplary embodiment of a paddle **520** that includes channels or holes according to the present disclosure. The paddle **520** includes a planar portion **524** that extends to a handle portion **528**. Rounded edges **252** are disposed along all exterior edges of the planar portion **524** and the handle portion **528**. The planar portion **524** includes a rigid portion that comprises a core **532** that includes multiple channels **536**. The channels **536** can be arranged in any of various patterns, without limitation, that are found to desirably enhance the performance of the paddle **520**.

**[0078]** Moreover, any one or more of the channels **536** can extend partially or all the way through the core **532**. In an embodiment, shown in FIG. **16**A, for example, a channel **536** extends all the way through the core **532**. In an embodiment, shown in FIG. **16**B, however, a first channel **540** and a second channel **544** are disposed on opposite sides of the planar portion **524**, leaving a relatively thin portion of the core **532** to separate the first and second channels **540**, **544**. In some embodiments, a rigid layer, such as the rigid layers **220** of FIG. **7**, can be disposed outside the core **532** to cover the channels **536**. The rigid layers **220** can comprise multilayer carbon fiber, fiberglass, or another similar material, without limitation. Thus, once the paddle **520** is wrapped in skin layers **168**, as described herein, the rigid layers will provide a flat paddle surface.

**[0079]** While the paddle and methods have been described in terms of particular variations and illustrative figures, those of ordinary skill in the art will recognize that the paddle is not limited to the variations or figures described. In addition, where methods and steps described above indicate certain events occurring in certain order, those of ordinary skill in the art will recognize that the ordering of certain steps may be modified and that such modifications are in accordance with the variations of the paddle. Additionally, certain of the steps may be performed concurrently in a parallel process, when possible, as well as performed sequentially as described above. To the extent there are variations of the paddle, which are within the spirit of the disclosure or equivalent to the paddle found in the claims, it is the intent that this patent will cover those variations as well. Therefore, the present disclosure is to be understood as not limited by the specific embodiments described herein, but only by scope of the appended claims.

What is claimed is:

**1**. A sports paddle, comprising:

a planar portion having opposing flat surfaces for striking a ball;

a core material having rounded edges;

a handle portion for grasping in a hand; and

a skin layer surrounding at least a majority of the planar portion.

**2**. The sports paddle of claim **1**, wherein the planar portion includes a shaped core material and a rigid portion.

**3**. The sports paddle of claim **2**, wherein the shaped core material comprises a window-like structure that provides a planar portion that exhibits a relatively greater degree of rigidity.

**4**. The sports paddle of claim **1**, wherein one or more weights are incorporated into the planar portion to increase the weight of the paddle.

**5**. The sports paddle of claim **4**, wherein any one or more of the one or more weights comprises a stabilizer that provides desired performance results.

**6**. The sports paddle of claim **5**, wherein any one or more of the multiple weights and the stabilizers comprise slugs of gel rubber for weighing the paddle and/or creating a dampening of vibration.

**7**. The sports paddle of claim **5**, wherein any one or more of the multiple weights and the stabilizers comprises a first weight and a second weight that are embedded on opposite sides of the core.

**8**. The sports paddle of claim **1**, wherein the planar portion comprises multiple stringers that are embedded in the core material.

**9**. The sports paddle of claim **8**, wherein the multiple stringers comprise strips of wood or plastic that are glued into the core material.

**10**. The sports paddle of claim **8**, wherein the multiple stringers comprise rubber, a foam, or another suitable material that provides a flexible planar portion.

**11**. The sports paddle of claim **8**, wherein one or more of the multiple stringers comprises a curved stringer that is embedded into the planar portion.

**12**. The sports paddle of claim **1**, wherein the planar portion comprises a rigid portion that includes a core material having one or more channels that are arranged in a pattern that desirably enhances the performance of the paddle.

**13**. The sports paddle of claim **12**, wherein any one or more of the one or more channels extends all the way through the core material.

**14**. The sports paddle of claim **12**, wherein any one or more of the one or more channels comprises a first channel and a second channel that are disposed on opposite sides of the planar portion, leaving a relatively thin portion of the core material disposed between the first channel and the second channel.

**15**. The sports paddle of claim **1**, wherein the planar portion includes an internal string core surrounded by structured foam layers.

**16**. The sports paddle of claim **15**, wherein the string core is strung on a rigid portion comprising carbon fiber, aluminum, or another similar material capable of supporting the string core.

**17**. The sports paddle of claim **15**, wherein a rigid layer is disposed outside each structured foam layer.

**18**. The sports paddle of claim **1**, wherein the planar portion includes a core material surrounded by an external string core that is strung around the exterior of a rigid portion.

**19**. The sports paddle of claim **18**, wherein the rigid portion comprises carbon fiber, aluminum, or another similar material.

**20**. The sports paddle of claim **18**, wherein the external string core and the rigid portion surround an interior volume that houses the core material.

* * * * *